

Financial Performance Analytics for Kenyan Commercial Banks

A Panel Data Analysis of the Impact of Macroeconomic Variables on Bank Profitability (2014–2024)

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Executive Summary

The profitability of commercial banks is influenced by both external economic conditions and internal operational performance. This project investigates how GDP growth, interest rates, inflation, asset quality, and management efficiency influence the financial performance of Kenyan commercial banks using quarterly data from 46 banks between 2014 and 2024. Python was used for data cleaning, exploratory analysis, panel regression modelling, and diagnostic testing. The findings indicate that GDP growth and interest rates positively influence profitability, while poor asset quality reduces profitability. Management efficiency was not statistically significant.

Business Problem

Commercial banks operate in a dynamic economic environment where changes in GDP growth, inflation, and interest rates affect lending, investment, and profitability. This study investigates how macroeconomic variables and bank-specific factors influence Return on Assets (ROA).

Project Objectives

- Examine the effect of GDP growth on ROA.
- Determine the effect of interest rates on profitability.
- Assess the impact of inflation.
- Evaluate asset quality as a control variable.
- Assess management efficiency as a control variable.

About the Dataset

Industry: Banking

Country: Kenya

Study Period: 2014–2024

Frequency: Quarterly

Banks: 46

Dependent Variable: ROA

Independent Variables: GDP Growth, Interest Rates, Inflation

Control Variables: Asset Quality, Management Efficiency

Methodology

Data Collection

Data Cleaning

Exploratory Data Analysis

Correlation Analysis

VIF Testing

Panel Regression (Pooled OLS, Fixed Effects, Random Effects)

Diagnostic Testing

Robust Standard Errors

Tools

Python, Pandas, NumPy, Statsmodels, Linearmodels, Matplotlib, OpenPyXL, Jupyter Notebook, Microsoft Excel

Key Findings

- GDP growth had a positive and statistically significant effect on ROA.
- Interest rates positively influenced profitability.
- Asset quality had a negative and statistically significant effect on profitability.
- Management efficiency was not statistically significant.
- Macroeconomic conditions and credit risk management remain key drivers of bank performance.

Business Implications

The findings demonstrate how macroeconomic indicators and internal bank-specific variables can support strategic planning, lending decisions, and risk management within commercial banks.

Conclusion

This project demonstrates my ability to collect, clean, analyse, model, and interpret financial data using Python and panel econometric techniques while communicating findings that support evidence-based decision-making.